

Project Design Phase

Solution Architecture

Date	5 July 2026
Team ID	
Project Name	Book A Doctor
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Solution Architecture Diagram (Textual Layout)

1. User Layer (Frontend)

- **Patients (Web/App):** Register, search doctors, view profiles, book appointments, receive reminders.
- **Doctors (Web/App):** Manage profile, set availability, view/manage appointments.

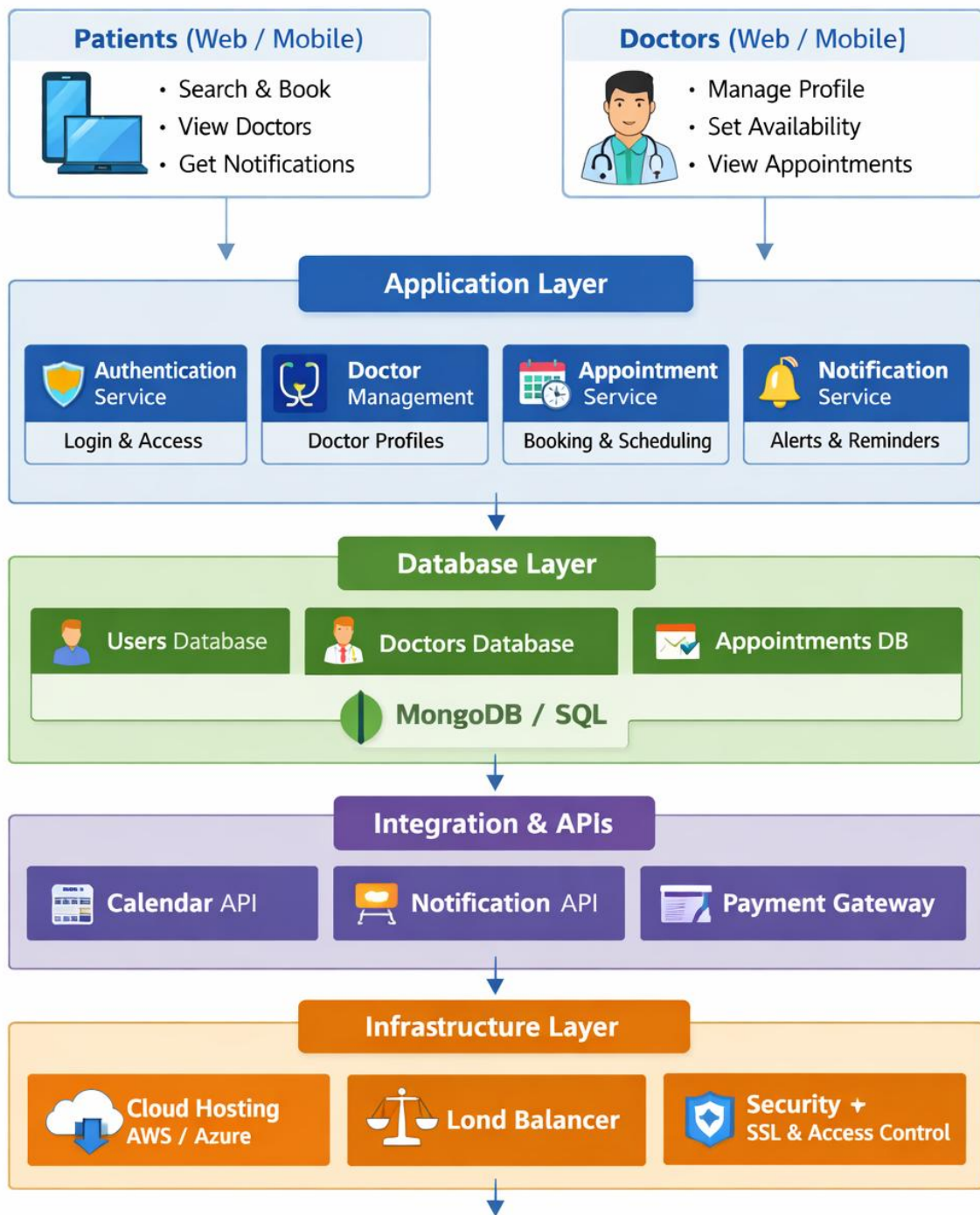
2. Application Layer (Backend Services)

- **Authentication Service:** Secure login for patients & doctors.
- **Doctor Management Service:** Handles doctor profiles, specialization, availability.
- **Appointment Management Service:** Booking, rescheduling, cancellations.
- **Notification Service:** Sends SMS/email/app reminders.
- **Payment Service (optional):** Online fee payments.

3. Database Layer (MongoDB Atlas)

- **Users Collection:** Patient details.
- **Doctors Collection:** Doctor profiles & schedules.
- **Appointments Collection:** Booking records.
- **Payments Collection (optional):** Transactions.

Doctor Appointment Booking System Architecture



- 1 Patient logs in → searches doctor → selects slot.
- 2 Doctor updates availability → confirms booking.
- 3 Appointment stored in database → notification sent to patient.
- 4 Doctor updates schedule → reflects in real-time.